VBF-T2R1NOFORCE-DS-001 - Cell Design Specification (F1)

Cell Design Specification - F1 (Grid Energy Storage Cell)

VBF-T2R1NOFORCE-DS-001 | Case: t2_r1_noforce | Date: 2026-08-25 | Base parameter set: Chen2020 (PyBaMM)

1. Basic specification

Item	Value	Source
Electrochemical system	NMC811 / graphite (Chen2020 teaching parameterization)	entry-0 meta, base Chen2020
Nominal capacity	5.0 Ah	parameter set `Nominal cell capacity [A.h]`
Verified 1C capacity (25 °C)	5.0282 Ah	`cell/r5_f1_1c_spme.json:capacity_ah`
Voltage window	2.5 - 4.2 V	parameter set `Upper/Lower voltage cut-off [V]`
Dimensions (H x W x T)	65 x 158 x 0.2008 mm	parameter set `Electrode height/width [m]`; `cell/r5_f1_energy.json:thickness_m`
Shell thickness	Not provided (no parameter)	-
Electrolyte formulation	EC/EMC + LiPF6 (task default); bulk solvent concentration 2636 mol/m^3; EC initial concentration 4541 mol/m^3; no additive applied (anode-side ceramic coating used instead)	parameter set `Bulk solvent concentration [mol.m-3]`, `EC initial concentration in electrolyte [mol.m-3]`
Cation transference number	0.2594	parameter set `Cation transference number`

2. Electrode and separator

Layer	Material	Thickness (µm)	Porosity	Active vol. fraction	Particle radius (µm)	Density (kg/m^3)
Positive electrode	NMC811	75.6	0.335	0.665	5.22	3262
Negative electrode	Graphite, ceramic-coated (ALD Al2O3/TiO2 class; k_SEI 4e-13 = x0.4)	85.2	**0.42 (design change)**	0.75	**3 (design change)**	1657
Separator	-	12	0.47	-	-	397
Positive current collector	Al	16	-	-	-	2700
Negative current collector	Cu	12	-	-	-	8960

- **N/P ratio (theoretical full-window areal-capacity ratio, deliverable formula: capacity density x thickness)**: 0.582 for F1 (baseline 0.753). Capacity density = c_max x F/3600 x active vol. fraction x (1-porosity) x thickness -> negative 32.91 Ah/m^2, positive 56.54 Ah/m^2 (computed from parameter set values above).
- **Practical (reversible) N/P**: ~ 0.88 - the design is deliberately anode-limited (measured cell capacity 5.0282 Ah below cathode practical capacity ~5.74 Ah from R1); the anode-limiting strategy, combined with porosity 0.42 + 3 µm particles, is what clears the 4C plating criterion.

3. Process design parameters

Parameter	Value	Formula / note
Positive areal density	164.0 g/m^2	75.6 µm x (1-0.335) x 3262 kg/m^3 (matches `r5_f1_energy.json:layer_kg_m2.positive_electrode`)
Negative areal density	81.9 g/m^2	85.2 µm x (1-0.42) x 1657 kg/m^3 (matches `r5_f1_energy.json:layer_kg_m2.negative_electrode`)
Positive compaction density	2.169 g/cm^3	3262 x 0.665 ÷ 1000
Negative compaction density	0.961 g/cm^3	1657 x 0.58 ÷ 1000
Electrolyte fill amount	8.23 g	pore volume 6.855 cm^3 x 1.2 g/cm^3 (literature value, annotated)
Formation recommendation	0.1C CC to 4.2 V, 25 °C, 2 cycles	design recommended value; actual production-line value requires tuning

4. Mass breakdown

Layer	Mass (g)	Source
Positive electrode (NMC811)	16.842	`r5_f1_energy.json:layer_kg_m2` x 0.1027 m^2
Negative electrode (coated graphite)	8.409	same
Positive current collector (Al)	4.437	same
Negative current collector (Cu)	11.042	same
Separator	0.259	same
Total (contract caliber, electrolyte excluded)	**40.990**	`r5_f1_energy.json:mass_kg`
Electrolyte (annotated, 1.2 g/cm^3 literature)	+8.23	§3

5. Performance verification

Metric	Criterion (entry 0)	Measured	Verdict
Energy density	>= 327.18 Wh/kg	435.44 Wh/kg	[PASS] PASS
SEI thickness @100 cyc	<= 500 nm	439.52 nm	[PASS] PASS

| SEI thickness @500 cyc | <= 550 nm | 738.43 nm | [FAIL] FAIL (documented unreachable - see final entry escalation) |

| -20 °C retention | >= 90 % | 99.45 % | [PASS] PASS |

| 4C charge plating | none (anode phi >= 0 V) | min anode potential +0.0071 V | [PASS] PASS |

| 4C max temperature (informational) | - | 359.92 K (86.8 °C, +41.8 K from 45 °C) | reported |

6. Design notes (changes vs baseline and why)

1. **Negative particle radius 5.86 -> 3 µm** (R2 V1): raises active surface area, drops anode overpotential at 4C - min anode potential -0.192 -> -0.138 V.
2. **Negative porosity 0.25 -> 0.42** (R3 V3/V6 -> R5 F1): the strongest single plating lever (0.35 -> -0.074 V; 0.40 -> -0.0017 V; 0.42 -> **+0.0071 V**).
3. **Ceramic-coated graphite, SEI kinetic rate constant 1e-12 -> 4e-13** (R3 C1, ALD Al2O3/TiO2 class): only lever that reduces SEI (-5.6 % @100 cyc, -4.2 % @500 cyc), bringing SEI@100 inside contract (439.5 <= 500).
4. Rejected directions (with evidence): N/P rebalance via thicker anode (R4 V7/V8 - SEI scales with anode thickness, both SEI contracts fail); electrolyte scalar overrides (would destroy Nyman2008 T-dependent transport); k_SEI beyond x0.4 (R5 F2 x0.1 -> 701.4 nm @500 cyc - saturating lever); system switch to OKane2022 (R5 S1 - graphite-only, z_sei=1, SEI worse: 626.8/1067.7 nm).